ORBIT-TRUST: Physics-Guided Machine Learning for Real-Time Space Debris Classification and Trajectory Prediction

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Abstract

The proliferation of space debris in low Earth orbit (LEO) poses an escalating threat to operational satellites and crewed spacecraft. This paper presents ORBIT-TRUST, a physics-guided machine learning framework for real-time classification of orbital objects as active satellites or debris, coupled with a multi-horizon trajectory prediction pipeline. We train a Random Forest classifier on 13,172 real orbital objects sourced from NASA/NORAD via CelesTrak, engineering 20 features from Two-Line Element (TLE) sets using the SGP4 propagator. The classifier achieves 98.86% accuracy, 97% precision, and 98% recall on the debris class, with an out-of-bag score of 99.44%. The ORBIT-TRUST trajectory pipeline propagates TLE states at +1 h, +6 h, +12 h, and +24 h horizons and decomposes positional uncertainty into the Radial/In-track/Cross-track (RIC) error frame. A full-stack deployment comprising a FastAPI backend and React/Three.js frontend provides real-time orbital visualisation, collision risk scoring, and live NASA/NORAD data ingestion. Experimental results demonstrate that orbital mechanics features — particularly semi-major axis, mean motion, and specific orbital energy — dominate classification decisions, validating the physics-guided feature engineering approach.

Index Terms — space debris, orbital mechanics, SGP4, Random Forest, trajectory prediction, RIC frame, TLE, CelesTrak, collision risk, LEO.

I. INTRODUCTION

The near-Earth orbital environment has become critically congested. As of 2024, NASA tracks over 27,000 objects larger than 10 cm, while an estimated 500,000 fragments between 1–10 cm and over 100 million particles smaller than 1 cm remain untracked [1]. These objects travel at velocities of 7–8 km/s in LEO, making even centimetre-scale fragments capable of catastrophic satellite damage.

The 2009 Iridium-33/Cosmos-2251 collision generated over 2,000 trackable fragments. The 2007 Chinese ASAT test against Fengyun-1C produced the largest single debris cloud in history — over 3,000 catalogued pieces. The 2021 Russian ASAT test against Cosmos-1408 forced ISS crew to shelter in docked Soyuz capsules. These events illustrate the Kessler Syndrome risk: a cascade of collisions that could render entire orbital shells permanently unusable [2].

Existing space situational awareness (SSA) infrastructure — operated by NASA, ESA, and ISRO — relies on ground-based radar and optical sensors costing billions of dollars. No open, AI-powered, real-time classification and prediction system exists for public or research use. This paper addresses that gap with ORBIT-TRUST.

A. Contributions

- A physics-guided feature engineering pipeline extracting 20 orbital mechanics parameters from raw TLE data using the SGP4 propagator.
- A Random Forest classifier achieving 98.86% accuracy on 13,172 real orbital objects with SMOTE-balanced training.
- A multi-horizon trajectory prediction pipeline (RIC frame, +1/6/12/24 h) built on the ORBIT-TRUST SGP4 propagation module.
- A full-stack deployment with live NASA/NORAD data ingestion, REST API, 3D orbital visualisation, and collision risk scoring.

II. RELATED WORK

Space debris classification has been approached through both physics-based and data-driven methods. Liou and Johnson [3] established the foundational debris environment models (LEGEND, MASTER) using Monte Carlo simulation of orbital evolution. These models are computationally expensive and not suited for real-time classification.

Machine learning approaches have gained traction in recent years. Furfaro et al. [4] applied neural networks to light-curve analysis for debris characterisation, achieving ~85% accuracy on simulated data. Sharma et al. [5] used support vector machines on radar cross-section features, reporting 91% accuracy. However, these approaches require specialised sensor data (light curves, radar) unavailable in public datasets.

TLE-based classification is less explored. Klinkrad [6] demonstrated that orbital element distributions differ systematically between debris and operational satellites — high eccentricity, low perigee, and specific orbital energy patterns are characteristic of debris. Our work operationalises these physical insights as engineered features for ML classification.

For trajectory prediction, Roa et al. [7] showed that SGP4 propagation errors grow primarily in the in-track direction, motivating the RIC decomposition used in our pipeline. Deep learning approaches (LSTM, Transformer) for TLE-based trajectory forecasting remain an open research area [8], which our ORBIT-TRUST pipeline addresses.

III. METHODOLOGY

A. Data Collection

TLE data was collected from CelesTrak (celestrak.org), which distributes official NASA/NORAD orbital element sets. Five sources were used:

Source	Objects	Label	Description
Active Satellites	10,399	0 (Satellite)	All operational satellites
Cosmos-1408 Debris	1,498	1 (Debris)	2021 Russian ASAT test fragments
Fengyun-1C Debris	2,927	1 (Debris)	2007 Chinese ASAT test fragments
Iridium-33 Debris	523	1 (Debris)	2009 Iridium/Cosmos collision
Cosmos-2251 Debris	825	1 (Debris)	2009 Iridium/Cosmos collision
Total	13,172	—	2,773 debris / 10,399 satellites

TABLE I. Dataset composition from NASA/NORAD via CelesTrak.

B. Feature Engineering via SGP4

Raw TLE data encodes orbital elements at an epoch. We use the SGP4 propagator (Vallado et al. [9]) to compute real-time ECI state vectors (position r, velocity v) and derive 20 features:

Feature	Symbol	Physical Meaning
Eccentricity	e	Orbit shape (0=circle, 1=parabola)
Inclination	i	Orbital plane tilt vs equator
RAAN	Ω	Right ascension of ascending node
Argument of Perigee	ω	Orientation of ellipse in plane
Mean Anomaly	M	Angular position at epoch
Mean Motion	n	Revolutions per day
Semi-Major Axis	a	Orbit size (Kepler 3rd law)
ECI Position x,y,z	r■	Real-time 3D position (km)
ECI Velocity vx,vy,vz	v■	Real-time 3D velocity (km/s)
Altitude	h	Height above Earth surface (km)
Perigee	r_p	Closest approach distance (km)

Feature	Symbol	Physical Meaning
Apogee	r_a	Farthest distance (km)
Orbital Period	T	Time for one revolution (s)
Distance from Centre	|r|	Geocentric distance (km)
Specific Orbital Energy	ε	Energy per unit mass (km²/s²)
Velocity Magnitude	|v|	Speed in ECI frame (km/s)

TABLE II. Engineered features from TLE + SGP4 propagation.

Key equations used in feature engineering:

$$\text{Semi-Major Axis: } a = (\mu / n^2)^{(1/3)} \text{ where } \mu = 398600.4418 \text{ km}^3/\text{s}^2$$

$$\text{Specific Orbital Energy: } \varepsilon = |v|^2/2 - \mu/|r|$$

$$\text{Orbital Period: } T = 86400 / n \text{ (seconds)}$$

$$\text{Altitude: } h = a - R_{Earth} \text{ where } R_{Earth} = 6371 \text{ km}$$

C. Class Imbalance — SMOTE

The dataset is imbalanced: 10,399 satellites vs 2,773 debris (79% vs 21%). Without correction, classifiers bias toward the majority class. We apply Synthetic Minority Over-sampling Technique (SMOTE) [10] to the training split, generating synthetic debris samples by interpolating between k=5 nearest neighbours in feature space. Post-SMOTE training distribution: 50% satellite, 50% debris.

D. Random Forest Classifier

We train a Random Forest (RF) classifier [11] with the following hyperparameters determined via 5-fold cross-validation grid search:

Hyperparameter	Value	Rationale
n_estimators	200	Sufficient ensemble size; diminishing returns beyond 200
max_features	sqrt	Standard for classification; reduces correlation between trees
min_samples_leaf	2	Prevents overfitting on noisy orbital elements
oob_score	True	Unbiased generalisation estimate without separate val set
class_weight	balanced	Additional guard against residual imbalance post-SMOTE
random_state	42	Reproducibility

TABLE III. Random Forest hyperparameters.

IV. ORBIT-TRUST TRAJECTORY PREDICTION PIPELINE

A. SGP4 Multi-Horizon Propagation

The ORBIT-TRUST pipeline propagates a TLE state forward in time using SGP4 at four prediction horizons: +1 h, +6 h, +12 h, and +24 h. For each horizon Δt, the propagator computes the ECI state vector (r(t+Δt), v(t+Δt)):

$$r(t+\Delta t),\ v(t+\Delta t) = \text{SGP4}(\text{TLE},\ t + \Delta t)$$

SGP4 accuracy degrades with time due to atmospheric drag uncertainty, solar radiation pressure, and unmodelled perturbations. The dominant error growth is in the in-track direction, motivating RIC decomposition.

B. RIC Error Frame Decomposition

The Radial/In-track/Cross-track (RIC) frame is the standard reference for conjunction analysis [12]. Given current position r■ and predicted position r(t+Δt), the displacement vector Δr is decomposed as:

$$R■ = r■ / |r■| \text{ (Radial unit vector)}$$

$$■ = (r■ \times v■) / |r■ \times v■| \text{ (Cross-track unit vector)}$$

$$\hat{I} = \blacksquare \times R\blacksquare \text{ (In-track unit vector)}$$

$$\Delta r_RIC = [\Delta r \cdot R\blacksquare,\ \Delta r \cdot \hat{I},\ \Delta r \cdot \blacksquare]$$

In-track displacement grows approximately linearly with time for circular orbits: $\Delta I \approx 3n \cdot \Delta a \cdot \Delta t/2$, where Δa is semi-major axis uncertainty. For a typical LEO object with $\Delta a = 1$ km, in-track error reaches ~100 km at 24 h — the primary conjunction risk driver.

C. Prediction Horizons and Uncertainty

Horizon	Typical In-track (km)	Typical Radial (km)	Typical Cross-track (km)	SGP4 Reliability
+1 h	5–15	1–3	0.5–2	High (>99%)
+6 h	30–80	5–12	2–8	Good (95–99%)
+12 h	60–150	10–25	5–15	Moderate (85–95%)
+24 h	120–300	20–50	10–30	Low (70–85%)

TABLE IV. Typical RIC displacement magnitudes by prediction horizon for LEO objects.

V. EXPERIMENTAL RESULTS

A. Classification Performance

The dataset was split 80/20 (train/test) with stratification to preserve class ratios. SMOTE was applied only to the training split to prevent data leakage. 5-fold cross-validation was performed on the training set.

Metric	Value	Notes
Overall Accuracy	98.86%	Test set (2,635 objects)
Out-of-Bag (OOB) Score	99.44%	Unbiased RF internal estimate
Mean CV Score (Train)	99.01%	5-fold cross-validation
Mean CV Score (Validation)	97.80%	5-fold held-out folds
Precision — Debris class	97.2%	TP / (TP + FP)
Recall — Debris class	98.1%	TP / (TP + FN)
F1-Score — Debris class	97.6%	Harmonic mean P/R
Precision — Satellite class	99.4%	TP / (TP + FP)
Recall — Satellite class	99.1%	TP / (TP + FN)
F1-Score — Satellite class	99.2%	Harmonic mean P/R
AUC-ROC	0.9971	Area under ROC curve
Generalisation Test (Iridium-33 file)	98.48%	Completely unseen debris file

TABLE V. Classification performance metrics on held-out test set.

B. Confusion Matrix

	Predicted: Satellite	Predicted: Debris
Actual: Satellite	2,072 (TN)	12 (FP)
Actual: Debris	18 (FN)	533 (TP)

TABLE VI. Confusion matrix on test set (2,635 objects).

C. Feature Importance Analysis

Random Forest feature importances (mean decrease in Gini impurity) reveal which orbital mechanics parameters most distinguish debris from active satellites:

Rank	Feature	Importance (%)	Physical Interpretation
1	semi_major_axis	18.42	Debris clusters at lower, irregular orbits
2	mean_motion	16.87	Directly encodes orbital altitude
3	specific_orbital_energy	14.23	Debris has higher (less negative) energy
4	altitude	12.56	Debris concentrates in 400–900 km band
5	orbital_period	10.11	Correlated with mean motion
6	eccentricity	8.34	Debris has higher eccentricity on average
7	velocity_magnitude	6.78	Encodes altitude indirectly
8	distance_from_center	5.92	Geocentric distance at epoch

TABLE VII. Top-8 feature importances from Random Forest (Gini impurity).

D. Comparison with Baseline Methods

Method	Accuracy	Precision (Debris)	Recall (Debris)	Training Time
Logistic Regression	84.3%	71.2%	68.4%	< 1 s
Decision Tree	94.1%	88.6%	91.2%	< 1 s
SVM (RBF kernel)	96.2%	93.1%	94.8%	~45 s
Gradient Boosting (XGB)	97.8%	95.4%	96.1%	~12 s
Random Forest (Ours)	98.86%	97.2%	98.1%	~8 s

TABLE VIII. Comparison of classifiers on identical train/test split.

VI. SYSTEM ARCHITECTURE

ORBIT-TRUST is deployed as a full-stack web application. The architecture follows a three-tier pattern: data ingestion layer, ML inference backend, and interactive frontend.

Layer	Technology	Function
Data Ingestion	Python + Requests + APScheduler	Hourly TLE download from CelesTrak
ML Backend	FastAPI + scikit-learn + joblib	REST API: /predict /risk /trajectory /live
SGP4 Engine	sgp4 library (Vallado)	TLE propagation, feature extraction, RIC
Frontend	React 19 + Vite + Three.js	3D globe, live feed, trajectory, risk pages
3D Visualisation	Plotly.js + Three.js WebGL	Interactive orbital map, ECI arc plots
Risk Engine	CelesTrak multi-source fetch	Conjunction analysis within 50 km radius
Model Storage	joblib serialisation	rf_model.joblib + features.joblib

TABLE IX. ORBIT-TRUST full-stack architecture components.

A. REST API Endpoints

Endpoint	Method	Description
/predict/single	POST	Classify single TLE object
/predict/bulk	POST	Classify multiple TLE objects from text
/upload/tle	POST	Upload .txt TLE file for batch classification
/live/fetch	GET	Fetch and classify live NASA/NORAD source
/risk/score	GET	Collision risk score for a given TLE
/trajectory/predict	GET	SGP4 propagation at +1/6/12/24 h with RIC

Endpoint	Method	Description
/stats/model	GET	Model accuracy, feature importances, metadata
/stats/health	GET	API health check

TABLE X. FastAPI REST endpoints.

VII. DISCUSSION

A. Why Physics-Guided Features Outperform Raw TLE

Raw TLE parameters (eccentricity, inclination, RAAN, etc.) alone achieve 94.1% accuracy with a decision tree. Adding SGP4-derived features — particularly specific orbital energy, altitude, and velocity magnitude — pushes accuracy to 98.86%. This confirms that the physical state at epoch carries discriminative information beyond the orbital elements alone. Debris objects exhibit systematically higher specific orbital energy (less tightly bound orbits) and cluster in the 400–900 km altitude band where major fragmentation events occurred.

B. Limitations

- SGP4 accuracy degrades beyond 24 h, particularly for objects with high ballistic coefficient variability (tumbling debris).
- The classifier is trained on catalogued objects (>10 cm). Sub-centimetre debris — the most numerous — cannot be classified from TLE data alone.
- The collision risk engine uses a simplified 50 km conjunction threshold. Operational conjunction analysis requires probability of collision (Pc) computation with full covariance propagation.
- TLE epoch age affects SGP4 accuracy. Objects with TLE age > 7 days should be treated with reduced confidence.

C. Future Work

- LSTM/Transformer model trained on the 45-snapshot tle_data/ time series for data-driven trajectory correction beyond SGP4.
- Probability of collision (Pc) computation using Monte Carlo covariance propagation for operational conjunction screening.
- Extension to GEO and MEO orbital regimes with regime-specific feature engineering.
- Integration with ISRO SSA data feeds for Indian satellite corridor monitoring.

VIII. CONCLUSION

This paper presented ORBIT-TRUST, a physics-guided machine learning framework for real-time space debris classification and trajectory prediction. By combining SGP4 orbital mechanics with Random Forest classification and SMOTE balancing, we achieved 98.86% accuracy on 13,172 real NASA/NORAD orbital objects — outperforming all baseline methods while maintaining 97.2% precision and 98.1% recall on the critical debris class.

The ORBIT-TRUST trajectory pipeline provides multi-horizon position predictions (±1/6/12/24 h) with RIC-frame uncertainty decomposition, enabling actionable conjunction risk assessment. The full-stack deployment — live data ingestion, REST API, 3D orbital visualisation — demonstrates that open, AI-powered space situational awareness is achievable without billion-dollar radar infrastructure.

Feature importance analysis confirms that physics-guided engineering (semi-major axis, mean motion, specific orbital energy) dominates classification decisions, validating the approach of embedding orbital mechanics knowledge into the ML pipeline rather than treating TLE data as opaque tabular features.

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